

Brayton Hall

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Data Scientist | Data Engineer

Experience in data acquisition and modeling, statistical analysis, machine learning, and natural language processing. With a background in mathematical logic and philosophy at Virginia Tech, I excel at **communicating insights on technical, high dimensional, and structured or unstructured data in a way that is interpretable and precise**. My experience with data science, and my graduate experience with scientific philosophy and statistical inference, teaching, and writing are ideal for helping data driven companies make **informed, optimal decisions**.

TECHNICAL SKILLS

Machine Learning, Applied Statistics, SQL, Python, Docker, PySpark, Hadoop, scikit-learn, NumPy, Pandas, Keras, Regression, Classification, Clustering, Seaborn, matplotlib, Git, NLTK, Gensim, Word2Vec, BeautifulSoup, spaCy, Heroku, Streamlit, React, HTML, CSS, Javascript

EMPLOYMENT HISTORY

Data Analyst, **PrimeNeuro**, Durham, North Carolina 10/2020 - 04/2021

- Used machine learning algorithms on fMRI brain-imaging data to predict emergent behavior, with the aim of early interventions for autism by creating behavioral classification models from high-dimensional data
- Cleaned, visualized, and modeled image and time-ordered data using AWS S3 and EC2, SQL, Docker, Python, Jupyter, and statistical methods such as principal component analysis to extract features from large data sets

Statistics Instructor, **The Learning Curve**, Charlotte, North Carolina 11/2017 - 08/2019

- Taught statistics and math for students at UNC Charlotte and in high school, with a focus on applied concepts such as confidence intervals, t-tests, hypothesis testing, and A/B testing
- Specialized in teaching linear algebra, probability theory, Bayesian statistics, experimental design, and linear regression

Graduate Teaching Assistant, **Virginia Tech**, Blacksburg, Virginia 08/2015 - 05/2017

- Lectured a 150-person seminar, created and maintained a grading database for professors, taught and attended seminars focused on logic, statistical inference, the ethics of science and technology, and the philosophy of natural science
- Organized class logistics with professors for about 60 students per semester, managed and organized the annual Virginia Tech Graduate Philosophy Conference, held office hours, and graded student papers and exams

TECHNICAL PROJECTS

Semantic Search Engine - [Github](#)

Created a 'search between the lines' app to search by connotations and misremembered quotes rather than exact fragments, inspired by the frustration of simple search engines on e-readers and websites

- Used Doc2Vec (a neural net package based on Word2Vec) in Gensim to vectorize 60,000 paragraphs from free ebooks on Project Gutenberg, and used them to train a model which can infer meaning based on cosine similarity
- Cleaned, organized, and visualized the data (1.2 million words) using Pandas and Seaborn, and stored the model and DataFrame on an S3 bucket with AWS
- Used a Docker container to deploy the app's front end on Heroku, which can be found on the project's Github page

COVID-19 Twitter Topic Modeling - [Github](#)

Using a combination of natural language processing packages to find intelligible topic clusters on Twitter

- Scraped 100,000 'coronavirus' tweets using Twint and preprocessed them with Gensim, NLTK, and spaCy
- Used the Python package vaderSentiment to analyze tweets for non-neutral sentiments
- Created a Latent Dirichlet Allocation (LDA) model with Gensim to discover that five distinct Twitter communities emerged: 'Trump', 'Economy', 'Employment', 'Public Health', and 'UK/Boris Johnson'

EDUCATION

Flatiron School, New York, NY 01/2020 - 05/2020

Immersive Data Science Program

- *Relevant experience:* Python, SQL, Docker, Machine Learning, Computer Science, Data Visualization

Virginia Tech, Blacksburg, Virginia 08/2015 - 05/2017

MA in Philosophy

- *Relevant coursework:* Metalogic and Number Theory, Philosophy of Science, Ethics, Philosophy of Language

UNC Chapel Hill, Chapel Hill, North Carolina 08/2009 - 05/2013

BS in Psychology

- *Relevant coursework:* Statistics, Decision Theory, Calc I, Calc II, Symbolic Logic, Physics, Cognitive Psychology